Report

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0.1 Denotation

- N_T: Number of text requests;
- N_I: Number of image requests;
- $T_1, T_2, \ldots$: GPU processing time of each text request;
- $I_1, I_2, \ldots$: GPU processing time of each image request;
- W: Total GPU workload within one minute.

Accordingly, the total GPU workload can be expressed as:

$$W = \sum_{j=1}^{N_T} T_j + \sum_{k=1}^{N_I} I_k. \tag{1}$$

0.2 Step 1

0.2.1 Drawing the Box Plots

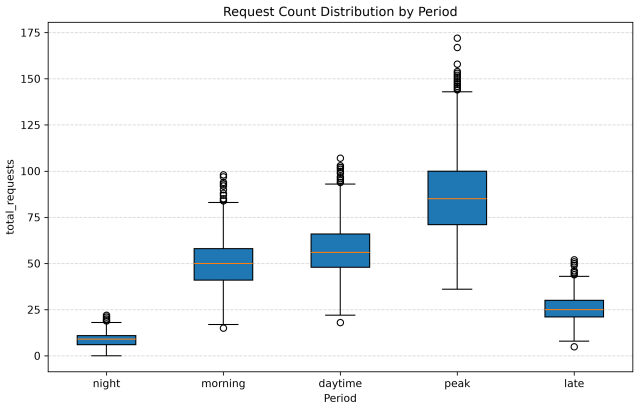


Figure 1: Box plots of text requests

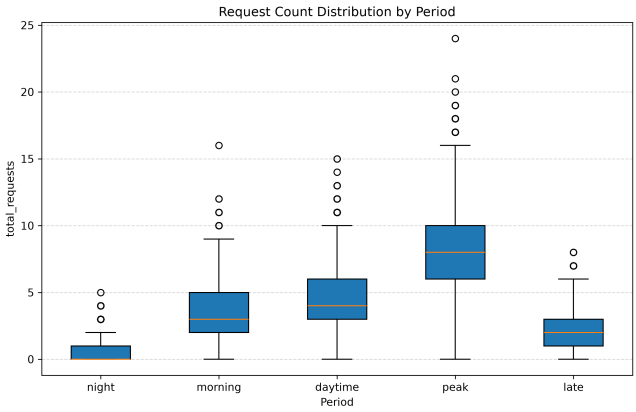


Figure 2: Box plots of image requests

The box plots for the two types of requests are presented in fig. 1 and fig. 2, respectively. As can be observed, an analysis limited exclusively to peak hours on weekdays is insufficient to provide a comprehensive representation of the actual trends in request volumes.

0.2.2 Establishing Poisson Distributions

We assume that N_t and N_i follow Poisson distributions, which can be expressed as follows:

$$P(N_t = k) = \frac{\lambda_t^k e^{-\lambda_t}}{k!},$$

$$P(N_i = k) = \frac{\lambda_i^k e^{-\lambda_i}}{k!}.$$

The parameters λ_t and λ_i can be estimated using the corresponding sample means:

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^{n} N_i.$$

The 95% confidence interval for the Poisson parameter λ can be approximated using the normal approximation:

$$CI_{95\%} = \left[\hat{\lambda} - 1.96\sqrt{\frac{\hat{\lambda}}{n}}, \hat{\lambda} + 1.96\sqrt{\frac{\hat{\lambda}}{n}} \right].$$

The estimation results are presented in table 1.

Table 1: Parameter Estimation for Poisson Distributions

Type	λ	95% (s)
Text	45.4513	[45.3583, 45.5444]
Image	3.6389	[3.6126, 3.6653]

0.2.3 Validating the Poisson Distributions

To evaluate the goodness of fit of the Poisson distributions, we conducted a χ^2 goodness-of-fit test, defined as:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i},$$

where O_i denotes the observed frequency of class i, and E_i denotes its corresponding expected frequency. The test results are presented in table 2.

Table 2: Results of the Poisson Goodness-of-Fit Tests

Type	χ^2	Conclusion
Text	8369296.14	Rejected
Image	72418.63	Rejected

0.2.4 Correlation Coefficient Between N_t and N_i

The correlation coefficient between N_t and N_i is defined as:

$$r = \frac{\sum_{i=1}^{n}(N_t - \bar{N})(N_i - \bar{N_i})}{\sqrt{\sum_{i=1}^{n}(N_t - \bar{N_t})^2}\sqrt{\sum_{i=1}^{n}(N_i - \bar{N_i})^2}}$$

. The estimated correlation coefficient is approximately 0.7851, indicating a strong positive correlation between text and image request volumes. This suggests that the two types of requests are likely influenced by common traffic patterns, potentially associated with multimodal requests.

0.3 Step 2

0.3.1 Estimation of W

We have:

$$E(W) = E(\sum_{j=1}^{N_T} T_j) + E(\sum_{k=1}^{N_I} I_k) = N_t E(T) + N_I E(I)$$

and

$$D(W) = D(\sum_{j=1}^{N_T} T_j) + D(\sum_{k=1}^{N_I} I_k) = N_t D(T) + N_I D(I)$$

. The estimated results are $E(W) = 61.8457$ and $D(W) = 51.3014$.

0.3.2 Assumption of Dependence Between N_T and N_I

The assumption that N_T and N_I do not subject to independence assumption will lead to $Cov(N_T, N_I) \neq 0$, which will cause error in

$$D(W) = N_t D(T) + N_I D(I) + Cov(T, I)$$

.

0.4 Step 3

0.4.1 title

0.4.2 Report on Service Time Data

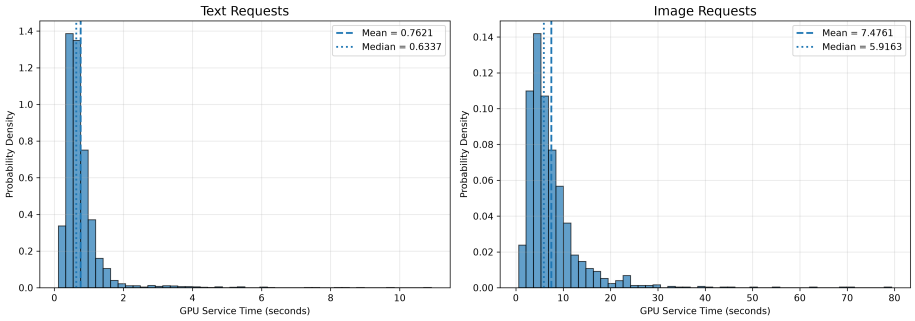


Figure 3: Histograms of the two types of requests

Table 3: Descriptive Statistics of Text and Image Request Durations (seconds)

Type	Size	Mean (s)	Std (s)	Median (s)	95% (s)	99% (s)
Text	4000	0.762147	0.628502	0.633650	1.464310	3.528130
Image	1600	7.476124	6.247750	5.916300	17.529665	29.749451

0.4.3 Analysis of the Effect of Cold Starts on Service Time

GPU system cold starts can result in substantial increases in response latency. The results presented in table 4 demonstrate that cold starts can considerably extend request processing times. According to the law of total probability:

$$P(T) = P(coldstart)P(t|coldstart) + P(hotstart)P(t|hotstart)$$

, cold starts can contribute to an increase in the overall mean response time and a heavier right tail in the response time distribution.

Table 4: Effect of Cold Starts on Response Time

Type	Cold Start(s)	Hot Start(s)
Text	3.6992	0.6884
Image	23.5168	6.6541

0.4.4 Establishing the Probability Model

A probability model can be constructed based on the observed service time distributions. In this report, the Gaussian distribution is adopted as the underlying

probability model. Specifically:

$$f(t_T) = \frac{1}{\sqrt{2\pi}\sigma_T} \exp\left(-\frac{(x-\mu_T)^2}{2\sigma_T^2}\right), f(t_I) = \frac{1}{\sqrt{2\pi}\sigma_I} \exp\left(-\frac{(x-\mu_I)^2}{2\sigma_I^2}\right)$$

0.4.5 Estimating the Model Parameters

The estimated model parameters are presented in table 5.

Table 5: Estimated Parameters of the Gaussian Distributions

μ_T	σ_T^2	μ_I	σ_I^2	$E(T^2)$	$E(I^2)$
0.7621	0.6285	7.4761	6.2477	0.9758	94.9024

0.5 Step 4

0.5.1 Estimation of W

We have:

$$E(W) = E(\sum_{j=1}^{N_T} T_j) + E(\sum_{k=1}^{N_I} I_k) = N_t E(T) + N_I E(I)$$

and

$$D(W) = D(\sum_{j=1}^{N_T} T_j) + D(\sum_{k=1}^{N_I} I_k) = N_t D(T) + N_I D(I)$$

. The estimated results are $E(W) = 61.8457$ and $D(W) = 51.3014$.

0.5.2 Estimation of m

To estimate m, which is subject to the condition $P(W > 48m)$, we can use the following approximation:

$$P(W > 48m) = P(\hat{W} > \frac{48m - E(W)}{\sqrt{D(W)}})$$

, where

$$\hat{W} = \frac{W - E(W)}{\sqrt{D(W)}}$$

and $\hat{W} \sim \mathcal{N}(0, 1)$. The resulting estimate is $m = 2.09$.

0.5.3 Rationale and Risks of Using the Central Limit Theorem

0.6 Step 5

0.6.1 title